

● MEDIUM

● **ADVANCED MATH**

Advanced Math

10

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1

ADVANCED MATH

$$f(\theta) = -0.28(\theta - 27)^2 + 880$$

An engineer wanted to identify the best angle for a cooling fan in an engine in order to get the greatest airflow. The engineer discovered that the function above models the airflow $f(\theta)$, in cubic feet per minute, as a function of the angle of the fan θ , in degrees. According to the model, what angle, in degrees, gives the greatest airflow?

- A. -0.28
- B. 0.28
- C. 27
- D. 880

Q2

GRID-IN

ADVANCED MATH

$$x^2 + x - 12 = 0$$

If a is a solution of the equation above and $a > 0$, what is the value of a ?

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

If $a = 4k + 5r$ and $b = 7k - 12r + 3$, which expression is equivalent to $a - b$?

A. $-3k + 17r + 3$

B. $-3k + 17r - 3$

C. $-3k - 7r - 3$

D. $-3k - 7r + 3$

Q4

GRID-IN

ADVANCED MATH

$$f(x) = x^5 + 9x + 17$$

For the given function f , the graph of $y = f(x)$ in the xy -plane passes through the point $(0, b)$, where b is a constant. What is the value of b ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$y = x + 1$$
$$y = x^2 + x$$

If (x,y) is a solution to the system of equations above, which of the following could be the value of x ?

- A. -1
- B. 0
- C. 2
- D. 3

Which expression is equivalent to $6x^8y^2 + 12x^2y^2$?

- A. $6x^2y^22x^6$
- B. $6x^2y^2x^4$
- C. $6x^2y^2x^6 + 2$
- D. $6x^2y^2x^4 + 2$

For the polynomial function f , the graph of $y = fx$ in the xy -plane passes through the points -50, 10, and 40. Which of the following must be a factor of $f(x)$?

A. $x + 1$

B. $x + 4$

C. $x - 1$

D. $x - 5$

If $4\sqrt{2x} = 16$, what is the value of $6x$?

A. 24

B. 48

C. 72

D. 96

$$gx = \frac{3}{5}x + \frac{7}{6}$$

$$hx = 6x - 5$$

The functions g and h are defined by the equations shown. Which expression is equivalent to $gx \cdot hx$?

A. $\frac{18x^2}{5} - \frac{35}{6}$

B. $\frac{18x^2}{5} + \frac{27x}{11} - \frac{35}{6}$

C. $\frac{18x^2}{5} - 4x - \frac{35}{6}$

D. $\frac{18x^2}{5} + 4x - \frac{35}{6}$

Immanuel purchased a certain rare coin on January 1. The function $f(x) = 651.03^x$, where $0 \leq x \leq 10$, gives the predicted value, in dollars, of the rare coin x years after Immanuel purchased it. What is the best interpretation of the statement “ $f(8)$ is approximately equal to 82” in this context?

- A. When the rare coin's predicted value is approximately 82 dollars, it is 8% greater than the predicted value, in dollars, on January 1 of the previous year.
- B. When the rare coin's predicted value is approximately 82 dollars, it is 8 times the predicted value, in dollars, on January 1 of the previous year.
- C. From the day Immanuel purchased the rare coin to 8 years after Immanuel purchased the coin, its predicted value increased by a total of approximately 82 dollars.
- D. 8 years after Immanuel purchased the rare coin, its predicted value is approximately 82 dollars.